GLOBAL AI HACKATHON | TEAM VISIONARIES

FreshCheck

Catch stale AI answers before they mislead you

AI often states old facts with full confidence. FreshCheck re-verifies every time-sensitive claim on the live web, shows the proof and fixes the answer. It works as a website, as an OpenAI-compatible layer in front of the Bifrost gateway, and as a tool inside Bifrost's MCP Gateway.

Tavily #1 Stale-answer detector

Bifrost #4 Gateway integration

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Dream . Innovate . Build

AI answers go stale, and still sound sure

Prices

“The latest iPhone is the iPhone 14.”

Versions

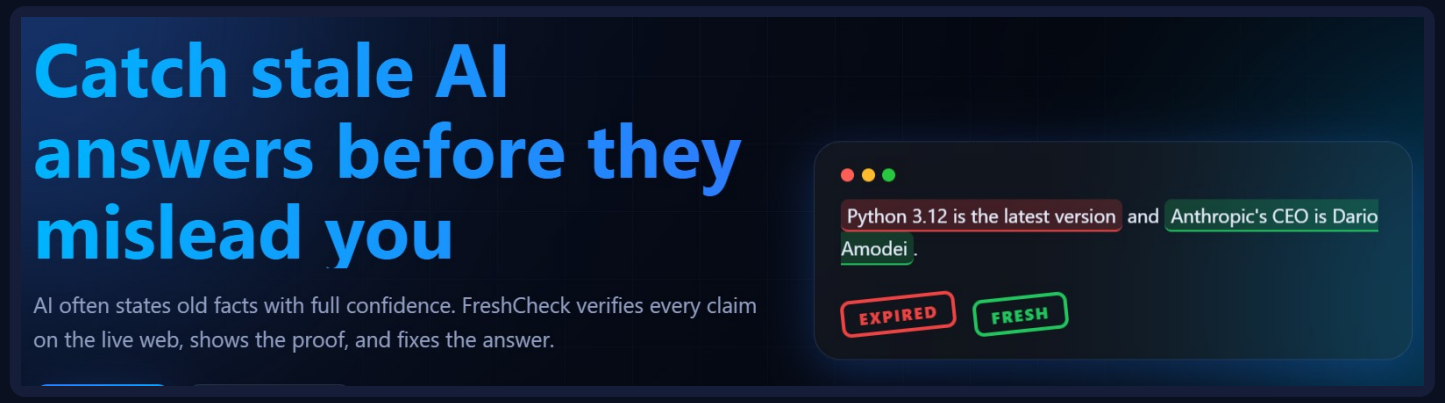
“Python 3.12 is the latest version.”

AI models

“ChatGPT runs on GPT-3.5.”

Laws

“The IPC is India’s main criminal law.”



Facts that change are the facts that hurt

These are real claims FreshCheck flagged in our tests. A user can’t tell which part of a confident answer is out of date.

FreshCheck verifies every fact that can change



STALE

Outdated. Shows the correct current value and a source link.



FRESH

Confirmed current by a live source, with the proof quote.



UNCLEAR

Not enough evidence, so it is flagged instead of guessed.

For every claim you get

- A verdict: stale, fresh or unclear
- A freshness score from 0 to 100
- The correct, current value
- An evidence quote and source link
- A best-before date
- A fixed version of the answer

Red = outdated Green = current Amber = unclear

Why FreshCheck is different

NOT ENOUGH

Plain AI answer

- Sounds sure, shows no proof
- Knowledge stops at its training date
- Cannot tell which facts have changed

PARTIAL

Normal web search

- Gives links, you do the reading
- Old pages often rank above new ones
- No verdict and no score

OUR APPROACH

FreshCheck

- Finds only facts that can change
- Recency probe looks for the update
- Verdict, score, proof quote and a fix

FreshCheck checks an answer after it is written, and shows proof for every claim.

How FreshCheck works



STEP 1

Input

Paste any AI answer, or ask a question.



STEP 2

Extract

Finds only facts that can change: prices, versions, leaders, laws.



STEP 3

Search

Tavily live search plus a recency probe that looks for updates.



STEP 4

Verify

Proof quote checked inside the source. Score from 0 to 100.



STEP 5

Fix

Highlights, correct value, source link and a one-click fix.



A verdict without proof is never trusted. The evidence quote must really appear inside the source page.

What you get in the app

Source proof

Evidence quote and source link, checked inside the page.

Explainable score

0 to 100 with a “Why this score?” breakdown.

Time machine

See how a fact changed over the years.

Best-before date

Each fact gets a shelf life: prices 30 days, laws 180.

Fix my answer

Rewrites the answer using the current facts.

Compare models

Which AI is the most outdated? Ask several at once.

Voice + languages

Voice chat and read aloud in English, Hindi, Marathi, Hinglish.

Share and export

Share link, PDF, copy or .txt. One-click demo traps.

Tavily #1: Stale-answer detector

What the problem statement asks, and what FreshCheck does

Takes any LLM answer	→	Paste text on the website, or call /v1/chat/completions with any model.
Flags claims likely to have gone stale	→	Extracts prices, versions, leaders and laws, and marks each STALE, FRESH or UNCLEAR.
Re-verifies against live sources	→	Tavily live search plus a recency probe. The proof quote is checked inside the source.
Freshness score per claim	→	A 0 to 100 score on every claim, with a “Why this score?” breakdown.

Bifrost #4: a feature for the gateway

MCP tool server

FreshCheck is registered in Bifrost's MCP Gateway as check_freshness, so models behind Bifrost can verify their own facts.

Freshness layer

An OpenAI-compatible endpoint in front of Bifrost. Every reply carries x-freshcheck score and status headers.

Compare via gateway

One question to several models, all through Bifrost, to show which AI is the most outdated.

Bifrost MCP Catalog: freshcheck is Healthy, 1/1 tools enabled, 1/1 auto-execute

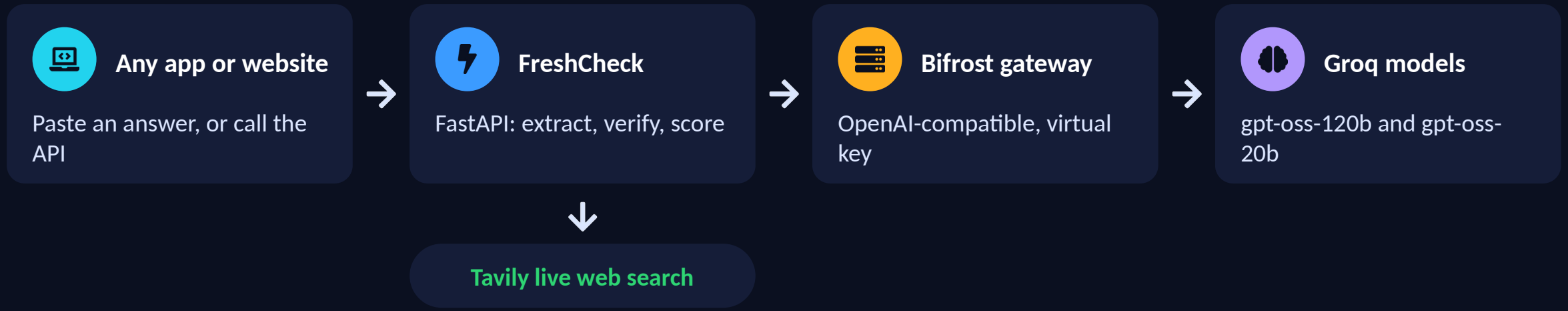
Name	Endpoint	Connection Type	Auth Type	Auth Scope	Code Mode	Access	Enabled Tools	Auto-execute Tools	State	
freshcheck	/mcp/freshcheck	HTTP	None	Shared	Disabled	1 VK	1/1	1/1	Healthy	•••

Bifrost MCP Logs: a model called check_freshness through the gateway and the call succeeded

Time	Tool Name	Source	Server	App	Latency	Cost	User	Team	
Nov 05 11:50:03 7 minutes ago	check_freshness Succeeded	MCP	freshcheck	Other	6,152ms Execution time	N/A	-	-	•••

Architecture: two ways into Bifrost

Flow 1 Freshness layer in front of Bifrost



Flow 2 Freshness tool inside Bifrost (MCP)



Every LLM call in both flows goes through Bifrost, so the gateway keeps control of keys, models and logs.

Drop-in for any app

1. Send a normal chat request

```
import requests

r = requests.post(
  "http://127.0.0.1:8000/v1/chat/completions",
  json={
    "model": "groq/openai/gpt-oss-120b",
    "messages": [{"role": "user",
      "content": "What is the latest Python?"}]
  },
  timeout=180,
)
print(r.headers["x-freshcheck-status"])
print(r.json()["freshcheck"]["claims"])
```

2. Get the answer + freshness info

```
HTTP 200
x-freshcheck-status: stale-found
x-freshcheck-score:  42
x-freshcheck-stale:  1

{
  "choices": [ { "message": { "content": "..." } } ],
  "freshcheck": {
    "score": 42, "fresh": 1, "stale": 1, "unclear": 0,
    "claims": [ {
      "claim": "Python 3.12 is the latest version",
      "verdict": "stale", "score": 3,
      "correct_value": "Python 3.15.0",
      "source_url": "https://www.python.org/downloads/",
      "evidence": "Python 3.15.0 Oct. 9, 2026 ..."
    } ]
  }
}
```

2. Get the answer plus freshness info

Challenges we faced, and how we solved them

1	**Old pages outrank new ones** Search often shows outdated pages first.	A recency probe: a second search aimed at updates and replacements, so the new fact appears in the results.
2	**AI can invent proof** A model may quote evidence that is not there.	The evidence quote must really appear inside the source page, or the verdict is not trusted.
3	**Live search credits are limited** Every check calls the Tavily API.	Search results and full checks are cached in files, so a repeated claim costs no new credits.
4	**Some claims cannot be proven** A guess would mislead the user.	Those claims are marked UNCLEAR instead of guessed, and every score shows its plus and minus points.

Technology used

Languages

- Python 3 (backend, evaluation)
- JavaScript, HTML, CSS (web app)
- Batch script (one-click start)
- JSON (file cache and stats)

Frameworks and libraries

- FastAPI and Uvicorn
- Pydantic
- OpenAI Python SDK (talks to Bifrost)
- tavily-python
- python-dotenv, requests

APIs and platforms

- Tavily Search API
- Bifrost AI gateway
- Groq models: gpt-oss-120b, gpt-oss-20b
- Model Context Protocol (MCP)
- Browser Web Speech API

Project structure and endpoints

```
freshcheck/
+-- server.py       web server, all endpoints
+-- extractor.py    finds facts that can go stale
+-- verifier.py     live search, verdict, score
+-- config.py       Bifrost client, Tavily, cache
+-- mcp_server.py   MCP tool: check_freshness
+-- index.html      website, 4 languages, voice
+-- eval.py         12-case accuracy test
+-- run.bat         starts Bifrost + FreshCheck
+-- test_*.py       gateway, Tavily, MCP tests
```

API endpoints

```
POST /check
POST /ask
POST /fix
POST /history
POST /compare
GET  /stats
POST /v1/chat/completions
POST /mcp
```

/v1/chat/completions is the gateway layer. /mcp is the Bifrost MCP tool.

Results and proof

12/12

internal test cases correct (9 stale, 2 fresh, 1 timeless)

4

languages: English, Hindi, Marathi, Hinglish

0-100

explainable freshness score on every claim



Dashboard: which kinds of facts go stale most often. In our test checks, pricing and versions were the most often outdated.



Compare models: gpt-oss-120b scored 30 and gpt-oss-20b scored 22. FreshCheck flagged the 20b answer, which named Python 3.13.0.

Demo walkthrough: try it in 60 seconds

1. **Pick a demo trap**
   One click loads a classic outdated AI answer.

2. **FreshCheck extracts the claims**
   Only facts that can change, like laws and versions.

3. **Live search and verify**
   Tavily search plus recency probe, proof checked.

4. **Read the verdict**
   STALE flag, correct value and source link.

5. **Fix it or compare models**
   Rewrite the answer, or rank several AIs.

REAL RESULT FROM OUR TEST

STALE

15/100

“Indian Penal Code (IPC) is the main criminal law in India”

CURRENT VALUE

Bharatiya Nyaya Sanhita, 2023 (effective July 1, 2024) replaced the IPC.

Proof: source link and evidence quote

Who uses it, and why it helps

Who will use it



AI app builders

Add a freshness check to any chatbot through the gateway.



Students and researchers

Check AI answers before using them in work.



Businesses and support teams

Avoid wrong prices, rules and versions.



Fact-checkers and everyday users

Voice and Hindi, Marathi, Hinglish support.

Benefits



Trust

Proof for every claim, not just a score.



Any model

Works with any model that runs through Bifrost.



Lower cost

Result and search cache save Tavily credits.



Explainable

Every score shows exactly why it was given.

Limits today, and what we build next

Honest limits today

Needs internet and API keys
Live search and the gateway need Tavily and Groq keys.

Small test set
12 internal cases so far, so a bigger benchmark is next.

Depends on what search finds
If no source proves a claim, it is marked UNCLEAR.

Runs locally for now
Start run.bat and open 127.0.0.1:8000.

What we build next

1. **Host it online**
A public link so anyone can try it without setup.

2. **Browser extension**
Check an AI answer right where you read it.

3. **Bigger benchmark**
Hundreds of claims, more languages and fact types.

4. **Auto re-check**
Re-verify a fact when its best-before date passes.

Thank you

FreshCheck: catch stale AI answers before they mislead you

github.com/yogeshx21/freshcheck

Run locally: start run.bat, then open 127.0.0.1:8000

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Global AI Hackathon | Tavily #1 + Bifrost #4